



# VICTORIAN CERTIFICATE OF EDUCATION

2011

NAME:

## MATHEMATICAL METHODS (CAS)

### Test 2 (CAS-FREE)

**Reading Time: 10 minutes**

**Writing time: 60 minutes**

### QUESTION AND ANSWER BOOK

| <i>Section</i> | <i>Number of Questions</i> | <i>Number of questions to be answered</i> | <i>Number of marks</i>  |
|----------------|----------------------------|-------------------------------------------|-------------------------|
| 1              | 9                          | 9                                         | Either 37, 38, 39 or 40 |

Test 2 consists of 9 short-answer questions.

In Question 1 and Question 6 there is a choice of which parts to do. The number of marks for this test will depend on the choice made.

Students are allowed to 'phone a friend', that is, ask a teacher for specific help if stuck on any question. The 'phone a friend' option can be used a maximum of 4 times. Each use will be recorded in the table below. Depending on the help given, there may be a 1 mark deduction when 'phone a friend' is used.

You must complete the test in the time allocated.

Students are NOT permitted to bring into the test: calculator, sheets of paper, books and/or white out liquid/tape.

Students are NOT permitted to bring mobile phones and/or any electronic communication devices into the test.

Phone a friend:

|                                                                                     |                                                                                     |                                                                                     |                                                                                     |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|  |  |  |  |
| Q                                                                                   | Q                                                                                   | Q                                                                                   | Q                                                                                   |

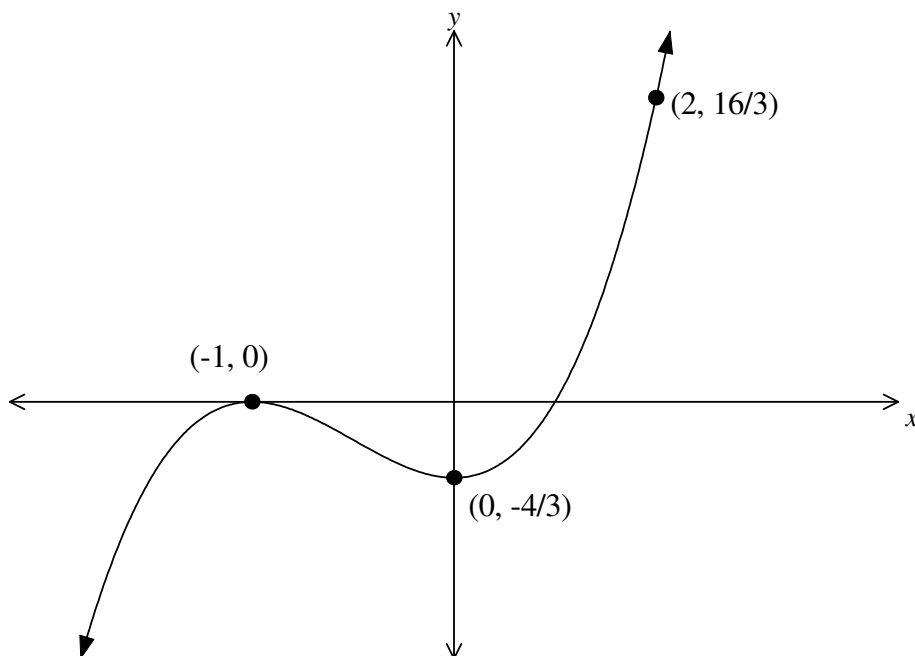
## Instructions

Answer all questions in the space provided.

In questions where more than one mark is available, appropriate working **must** be shown.

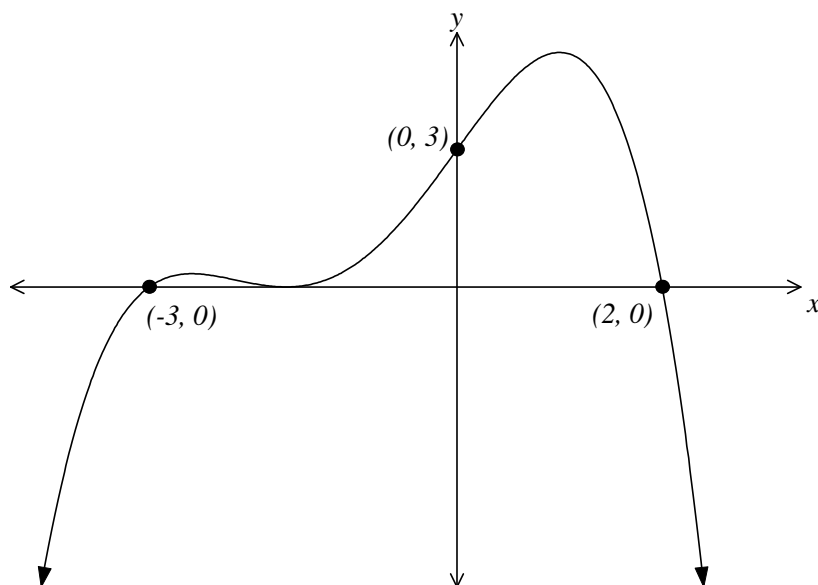
### Question 1

- a.** The graph of a cubic function is shown below. Determine a rule for this function.

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.



- c. The graph of a quartic function with three  $x$ -intercepts and a maximum turning point at  $\left(1, \frac{128}{25}\right)$  is shown below. Determine a rule for this function.

[illegible]

4 marks

Total either 5 or 7 marks











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4 marks

**Question 7**

Let  $f: \mathbb{R} \rightarrow \mathbb{R}$ ,  $f(x) = 1 - e^{-2x}$ .

- a.** Find the rule and domain of the inverse function  $f^{-1}$ .

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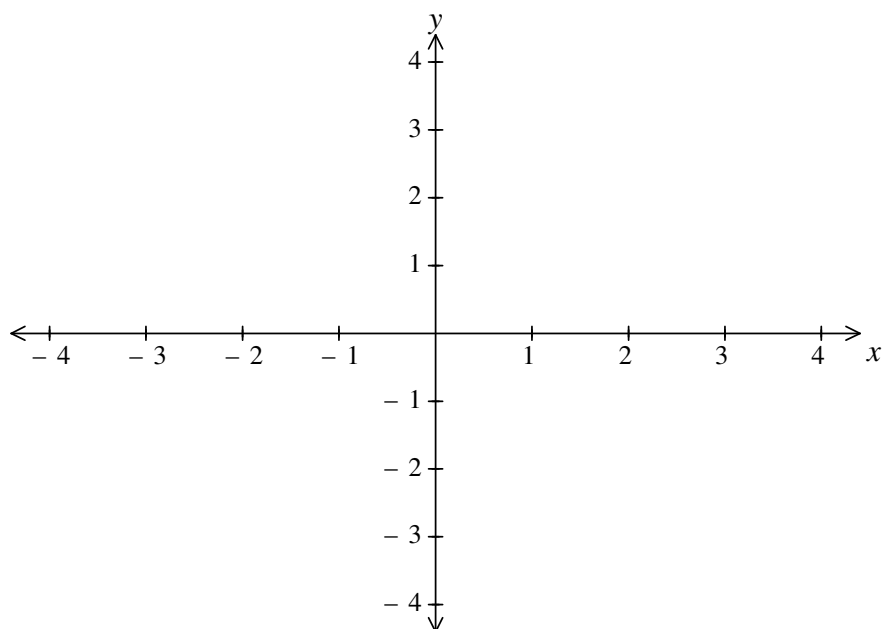
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2 marks

- b.** On the axes provided, sketch the graph of  $y = f(f^{-1}(x))$  for its maximal domain.



1 mark

c. Find  $y = f(-f^{-1}(-2x))$  in the form  $\frac{ax}{bx+c}$  where  $a$ ,  $b$  and  $c$  are real constants.

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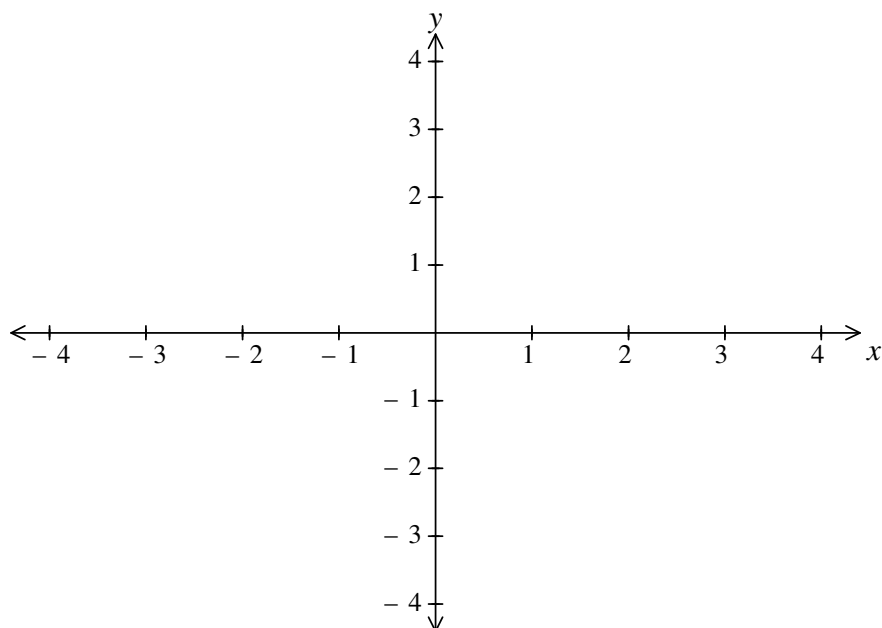
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3 marks  
Total 6 marks

**Question 8**

- a. On the axes provided, sketch the graph of  $y = \log_e(2 - 3x) + 1$ . Label the exact coordinates of all axes intercepts and label all asymptotes with their equation.



2 marks

Let  $f : D \rightarrow R$ ,  $f(x) = \log_e(|2 - 3x|) + 1$ .

- b. Find the maximal domain  $D$  of  $f$ .

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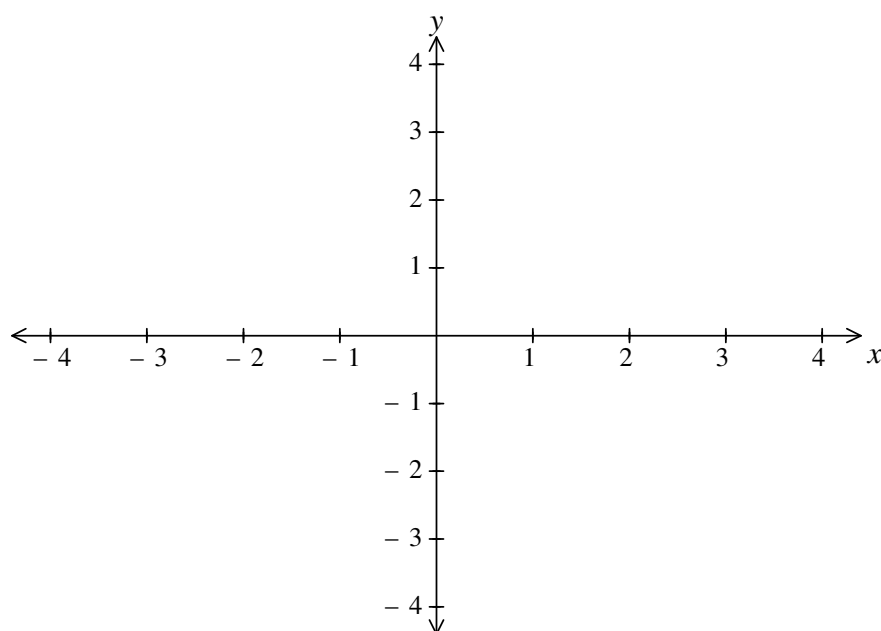
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1 mark

- c. On the axes provided, sketch the graph of  $y = f(x)$  for its maximal domain. Label the exact coordinates of all axes intercepts and label all asymptotes with their equation.



1 mark

Total 4 marks

**Question 9**

**a.** Let  $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = (x-3)^2(1-x)^2 + 1$ .

**i.** Find the derivative function  $f'(x)$  in factorised form.

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**ii.** Find the maximal subset of  $\mathbb{R}$  for which the gradient of  $f$  is negative.

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**b.** Let  $g : \mathbb{R} \rightarrow \mathbb{R}, g(x) = |8x - x^3|$ .

**i.** Complete the following to specify  $g'(x)$  as a hybrid function over the maximal domain of  $g'$ :

$$g'(x) = \begin{cases} \text{_____} & \text{for } x \in \text{_____} \\ \text{_____} & \text{for } x \in \text{_____} \end{cases}$$

**Working space:**

**ii.** Find the maximal subset of  $\mathbb{R}$  for which the gradient of  $g$  is positive.

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2 + 1 = 3 marks

Total 6 marks

**End of test**